

To run and test the application cleanly on another Windows machine with NVIDIA GPU acceleration, use the following requirements.txt manifest and setup guide.

1. requirements.txt

Create a file named requirements.txt in your project root folder and paste the following content:

```
Plaintext
# PyTorch Core (CUDA 12.1 builds must be installed via PyTorch index URL)
torch==2.5.1
torchvision==0.20.1
# Fine-Tuning, Quantization & Transformers
transformers>=4.45.0
peft>=0.12.0
bitsandbytes>=0.43.0
accelerate>=0.33.0
qwen-vl-utils>=0.0.8
# Computer Vision, Media Processing & Excel
opencv-python-headless
av
pillow<12
openpyxl
# UI & Core Utilities
requests
```

2. Machine Setup & Installation Steps

Execute these steps on the target PC to establish the environment:

Prerequisites

- **Python 3.12** installed (ensure the "Add python.exe to PATH" option was checked during installation).
- **NVIDIA Drivers** installed and up to date.

Installation Sequence

1. **Open Command Prompt / PowerShell** in the project directory:

```
DOS
cd path\to\ChatBot V3.0
```

2. **Create and Activate the Virtual Environment:**

```
DOS
py -3.12 -m venv venv
venv\Scripts\activate
```

3. **Install PyTorch with CUDA 12.1 Support:**

```
DOS
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121
```

4. **Install Remaining Requirements:**

```
DOS
pip install -r requirements.txt
```

5. **Verify GPU Availability:**

DOS

```
python -c "import torch; print('CUDA Available:', torch.cuda.is_available()); print('Device:', torch.cuda.get_device_name(0) if torch.cuda.is_available() else 'None')"
```

(Ensure it prints CUDA Available: True and names the NVIDIA GPU).

6. **Run the Application:**

DOS

```
python app.py
```